THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 7: **Proportional Reasoning-1** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 2 : 3	Divide both terms by 6
2	**(b)** 12	Multiply both terms by 4
3	**(c)** $2:3$ and $8:12$	Product of extremes $= 2 \times 12$ equals product of means $= 3 \times 8$
4	**(a)** 1 : 4	Convert to the same unit first
5	**(d)** ₹ 250	One part $= 400 \div 8 = 50$
6	**(b)** ₹ 8	$48 \div 6$
7	**(c)** ₹ 210	Find the cost of 1 book first
8	**(b)** 4 : 7	Divide both terms by 5
9	**(b)** 30	Multiply both terms by 6
10	**(c)** $4:5$ and $12:15$	Product of extremes $= 4 \times 15$ equals product of means $= 5 \times 12$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	2 m = 200 cm (1); ratio $= 1:5$ (1).	1+1
12	Both terms of the first ratio are multiplied by 3 (1); $x = 27$ (1).	1+1
13	Cost of 1 notebook $= 91 \div 7 = 13$ (1); cost of 12 $=$ ₹156 (1).	1+1
14	3 m = 300 cm (1); ratio $= 1:4$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Total parts 8 (1); one part ₹ 150 (1); shares ₹ 450 and ₹ 750 (1).	1+1+1
16	Product of extremes 3x20$= 60$ (1); product of means 4x15$= 60$ (1); equal, so in proportion (1).	1+1+1
17	Cost per kg $= 225 \div 5 = 45$ (1); cost of 8 kg $= 45 \times 8$ (1); $=$ ₹360 (1).	1+1+1
	OR: Total parts 7 (1); one part ₹ 120 (1); shares ₹ 240 and ₹ 600 (1).	1+1+1
18	Product of extremes 5x30$= 150$ (1); product of means 8x20$= 160$ (1); not equal, so not in proportion (1).	1+1+1
	OR: Cost per kg $= 180 \div 4 = 45$ (1); cost of 7 kg $= 45 \times 7$ (1); $=$ ₹315 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Total parts 8, one part $= 6$ (2); numbers 18 and 30 (2); difference 12 (1).	2+2+1
	OR: Cost of 1 pen ₹ 15 (1); cost of 10 pens ₹ 150 (2); pens for ₹ 210: 14 (2).	1+2+2
20	Total parts 9, one part $= 8$ (2); numbers 16 and 56 (2); difference 40 (1).	2+2+1
	OR: Cost of 1 pen ₹ 15 (1); cost of 15 pens ₹ 225 (2); pens for ₹ 135: 9 (2).	1+2+2

Mistakes to avoid:

- Forming a ratio without converting units: 50 cm : 2 m is $50:200 = 1:4$, not $25:1$.
- Writing the ratio in the wrong order: the ratio of A to B is not the same as B to A.
- Dividing a total in the ratio $3:5$ by dividing it by 3 and 5 separately; first add the parts ($3+5=8$) and divide by 8.
- Adding the same number to both terms to get an equivalent ratio ($2:3 \neq 4:5$); equivalent ratios are got by multiplying or dividing.
- Comparing ratios by only looking at the first terms; convert to fractions with the same denominator.
- Mixing up which quantities go together in the unitary method (e.g. dividing cost by the wrong number of items).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 7: **Proportional Reasoning-1** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 1 : 4	Convert to the same unit first
2	**(b)** ₹ 400	One part $= 720 \div 9 = 80$
3	**(b)** ₹ 9	$72 \div 8$
4	**(c)** ₹ 150	Find the cost of 1 book first
5	**(c)** 2 : 3	Divide both terms by 7
6	**(c)** 21	Multiply both terms by 3
7	**(d)** $3 : 7$ and $9 : 21$	Product of extremes $= 3 \times 21$ equals product of means $= 7 \times 9$
8	**(a)** 1 : 4	Convert to the same unit first
9	**(d)** ₹ 700	One part $= 900 \div 9 = 100$
10	**(b)** ₹ 9	$108 \div 12$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Both terms of the first ratio are multiplied by 6 (1); $x = 66$ (1).	1+1
12	Cost of 1 notebook $= 135 \div 9 = 15$ (1); cost of 4 $=$ ₹60 (1).	1+1
13	5 m = 500 cm (1); ratio $= 3 : 25$ (1).	1+1
14	Both terms of the first ratio are multiplied by 4 (1); $x = 8$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Total parts 9 (1); one part ₹ 150 (1); shares ₹ 600 and ₹ 750 (1).	1+1+1
16	Product of extremes 6x21$= 126$ (1); product of means 9x14$= 126$ (1); equal, so in proportion (1).	1+1+1
17	Cost per kg $= 270 \div 6 = 45$ (1); cost of 10 kg $= 45 \times 10$ (1); $=$ ₹450 (1).	1+1+1
	OR: Total parts 12 (1); one part ₹ 80 (1); shares ₹ 400 and ₹ 560 (1).	1+1+1
18	Product of extremes 4x22$= 88$ (1); product of means 7x12$= 84$ (1); not equal, so not in proportion (1).	1+1+1
	OR: Cost per kg $= 126 \div 3 = 42$ (1); cost of 11 kg $= 42 \times 11$ (1); $=$ ₹462 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Total parts 13, one part $= 5$ (2); numbers 20 and 45 (2); difference 25 (1).	2+2+1
	OR: Cost of 1 pen ₹ 20 (1); cost of 12 pens ₹ 240 (2); pens for ₹ 220: 11 (2).	1+2+2
20	Total parts 11, one part $= 12$ (2); numbers 60 and 72 (2); difference 12 (1).	2+2+1
	OR: Cost of 1 pen ₹ 15 (1); cost of 4 pens ₹ 60 (2); pens for ₹ 195: 13 (2).	1+2+2

Mistakes to avoid:

- Forming a ratio without converting units: 50 cm : 2 m is $50 : 200 = 1 : 4$, not $25 : 1$.
- Writing the ratio in the wrong order: the ratio of A to B is not the same as B to A.
- Dividing a total in the ratio $3 : 5$ by dividing it by 3 and 5 separately; first add the parts ($3 + 5 = 8$) and divide by 8.
- Adding the same number to both terms to get an equivalent ratio ($2 : 3 \neq 4 : 5$); equivalent ratios are got by multiplying or dividing.
- Comparing ratios by only looking at the first terms; convert to fractions with the same denominator.
- Mixing up which quantities go together in the unitary method (e.g. dividing cost by the wrong number of items).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 7: **Proportional Reasoning-1** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** ₹ 320	Find the cost of 1 book first
2	**(d)** 3 : 5	Divide both terms by 8
3	**(c)** 30	Multiply both terms by 5
4	**(b)** $5 : 6$ and $20 : 24$	Product of extremes $= 5 \times 24$ equals product of means $= 6 \times 20$
5	**(c)** 1 : 4	Convert to the same unit first
6	**(d)** ₹ 360	One part $= 660 \div 11 = 60$
7	**(a)** ₹ 13	$65 \div 5$
8	**(a)** ₹ 225	Find the cost of 1 book first
9	**(a)** 3 : 2	Divide both terms by 15
10	**(d)** 56	Multiply both terms by 7

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Cost of 1 notebook $= 114 \div 6 = 19$ (1); cost of 11 $=$ ₹209 (1).	1+1
12	4 m = 400 cm (1); ratio $= 9 : 40$ (1).	1+1
13	Both terms of the first ratio are multiplied by 5 (1); $x = 50$ (1).	1+1
14	Cost of 1 notebook $= 96 \div 8 = 12$ (1); cost of 15 $=$ ₹180 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Total parts 9 (1); one part ₹ 120 (1); shares ₹ 840 and ₹ 240 (1).	1+1+1
16	Product of extremes 8x15$=$ 120 (1); product of means 12x10$=$ 120 (1); equal, so in proportion (1).	1+1+1
17	Cost per kg $= 360 \div 8 = 45$ (1); cost of 5 kg $= 45 \times 5$ (1); $=$ ₹225 (1).	1+1+1
	OR: Total parts 11 (1); one part ₹ 70 (1); shares ₹ 210 and ₹ 560 (1).	1+1+1
18	Product of extremes 9x16$=$ 144 (1); product of means 5x27$=$ 135 (1); not equal, so not in proportion (1).	1+1+1
	OR: Cost per kg $= 297 \div 9 = 33$ (1); cost of 6 kg $= 33 \times 6$ (1); $=$ ₹198 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Total parts 10, one part $= 4$ (2); numbers 28 and 12 (2); difference 16 (1).	2+2+1
	OR: Cost of 1 pen ₹ 15 (1); cost of 20 pens ₹ 300 (2); pens for ₹ 90: 6 (2).	1+2+2
20	Total parts 13, one part $= 3$ (2); numbers 24 and 15 (2); difference 9 (1).	2+2+1
	OR: Cost of 1 pen ₹ 18 (1); cost of 9 pens ₹ 162 (2); pens for ₹ 270: 15 (2).	1+2+2

Mistakes to avoid:

- Forming a ratio without converting units: 50 cm : 2 m is $50 : 200 = 1 : 4$, not $25 : 1$.
- Writing the ratio in the wrong order: the ratio of A to B is not the same as B to A.
- Dividing a total in the ratio $3 : 5$ by dividing it by 3 and 5 separately; first add the parts ($3 + 5 = 8$) and divide by 8.
- Adding the same number to both terms to get an equivalent ratio ($2 : 3 \neq 4 : 5$); equivalent ratios are got by multiplying or dividing.
- Comparing ratios by only looking at the first terms; convert to fractions with the same denominator.
- Mixing up which quantities go together in the unitary method (e.g. dividing cost by the wrong number of items).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.